

Linux Basics Lab — Step-by-step with screenshots

1. Creating and Renaming Files / Directories

Goal: Create `test_dir`, add `example.txt`, then rename to `renamed_example.txt`.

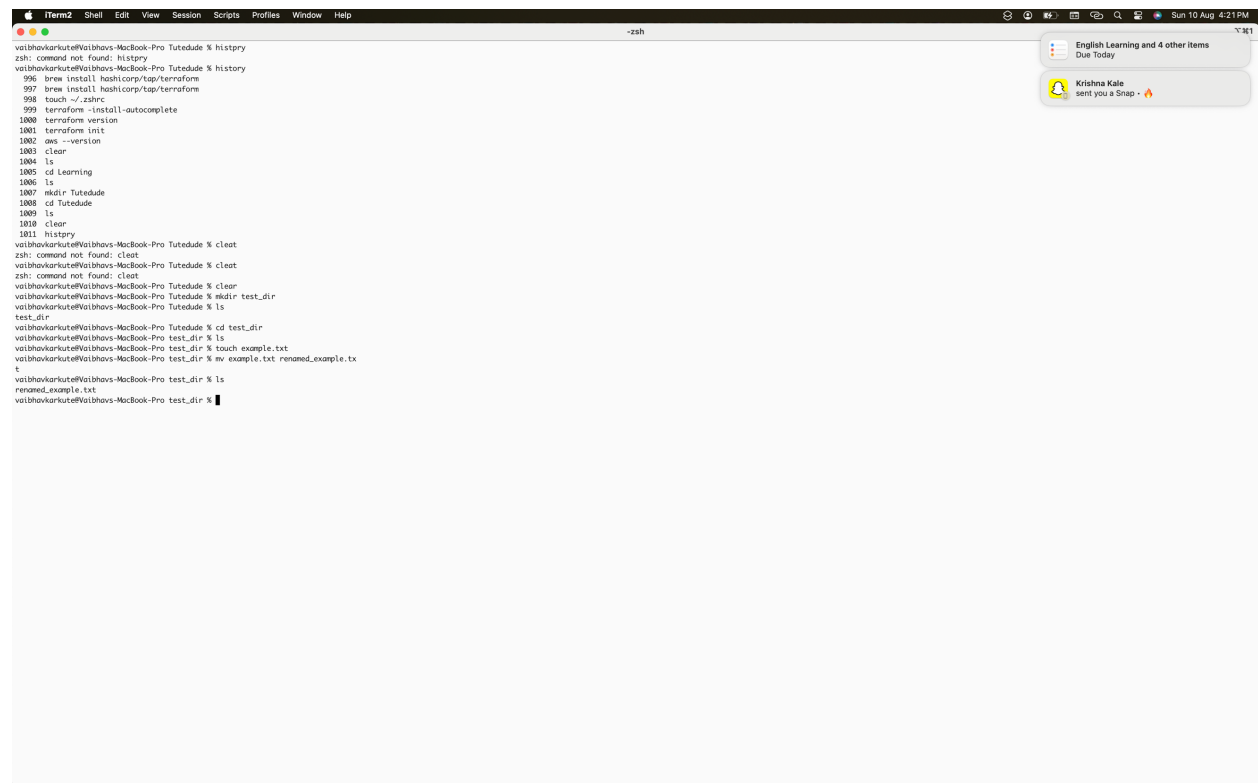
Commands:

```
mkdir test_dir
cd test_dir
touch example.txt      # create an empty file
ls                     # list files to confirm
mv example.txt renamed_example.txt
ls                     # verify rename
```

What this does:

- `mkdir` makes a directory. `touch` creates an empty file. `mv` renames or moves files. `ls` lists contents.

Screenshot (evidence):



The screenshot shows a terminal window titled "iTerm2" with a menu bar (Shell, Edit, View, Session, Scripts, Profiles, Window, Help) and a status bar (Sun 10 Aug 4:21 PM). The terminal output shows a sequence of commands and their results:

```
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % history
zsh: command not found: history
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % history
996 brew install hashicorp/tap/terraform
997 brew install hashicorp/tap/terraform
998 touch ~/.zshrc
999 terraform --install-autocomplete
1000 terraform version
1001 terraform init
1002 aws --version
1003 clear
1004 ls
1005 cd Learning
1006 ls
1007 mkdir Tutedude
1008 cd Tutedude
1009 ls
1010 clear
1011 history
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % cleat
zsh: command not found: cleat
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % cleat
zsh: command not found: cleat
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % clear
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % mkdir test_dir
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % ls
test_dir
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % cd test_dir
vaibhavkarkute@Vaibhavs-MacBook-Pro test_dir % ls
vaibhavkarkute@Vaibhavs-MacBook-Pro test_dir % touch example.txt
vaibhavkarkute@Vaibhavs-MacBook-Pro test_dir % mv example.txt renamed_example.tx
t
vaibhavkarkute@Vaibhavs-MacBook-Pro test_dir % ls
renamed_example.txt
vaibhavkarkute@Vaibhavs-MacBook-Pro test_dir %
```

On the right side of the terminal window, there are two notification cards: "English Learning and 4 other items Due Today" and "Krishna Kale sent you a Snap".

2. Viewing File Contents

Goal: View /etc/passwd and display first/last 5 lines.

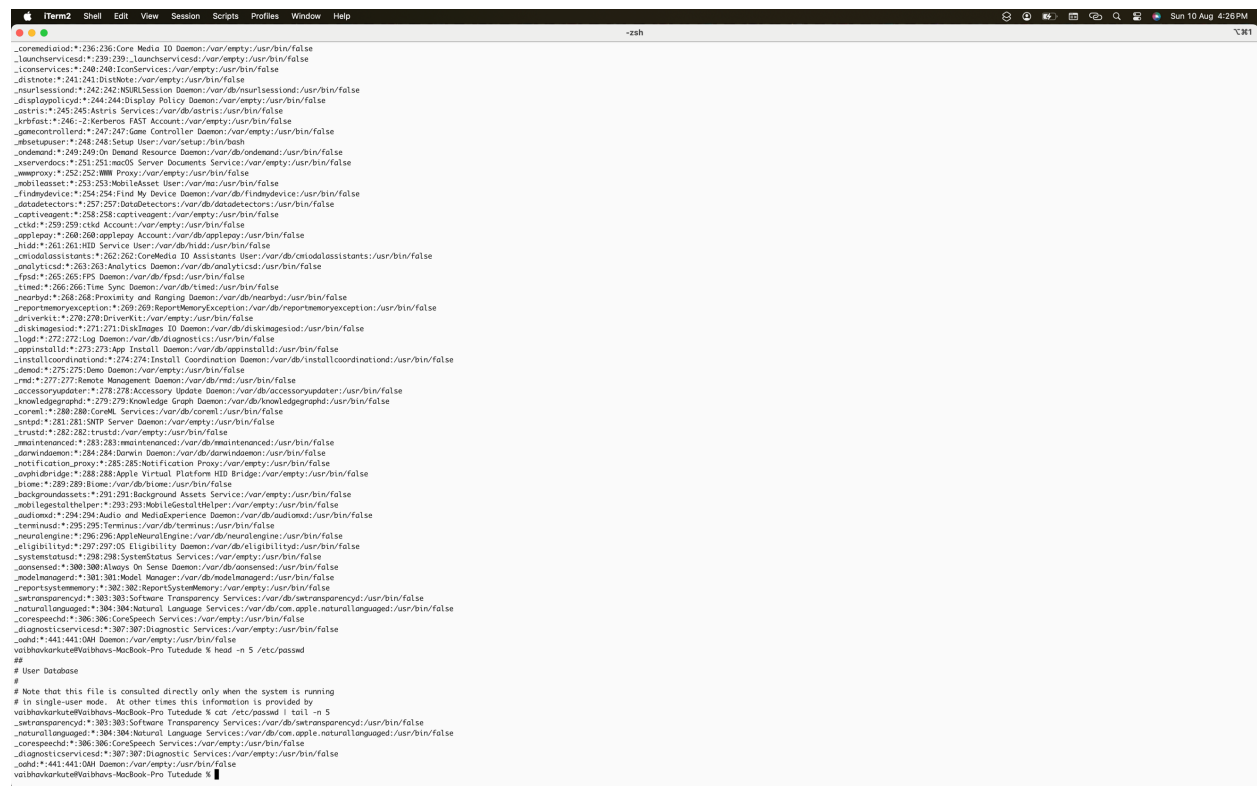
Commands:

```
cat /etc/passwd      # show full file (may be long)
head -n 5 /etc/passwd # first 5 lines
tail -n 5 /etc/passwd # last 5 lines
```

What this shows:

- /etc/passwd contains local user account information. Typical fields:
username:passwd:UID:GID:comment:home:shell.

Screenshot (evidence):



```
_coremediatd:*:236:236:Core Media IO Daemon:/var/empty:/usr/bin/false
_launchservicesd:*:239:239:_launchservicesd:/var/empty:/usr/bin/false
_iconservices:*:240:240:IconServices:/var/empty:/usr/bin/false
_distnote:*:241:241:DistNote:/var/empty:/usr/bin/false
_nsurlsessiond:*:242:242:NSURLSession Daemon:/var/db/nsurlsessiond:/usr/bin/false
_displayspolicyd:*:244:244:Display Policy Daemon:/var/empty:/usr/bin/false
_astris:*:245:245:Astris Services:/var/db/astis:/usr/bin/false
_krbfast:*:246:-2:Kerberos FAST Account:/var/empty:/usr/bin/false
_genecontrol:*:247:247:Gene Controller Daemon:/var/empty:/usr/bin/false
_absetupuser:*:248:248:Setup User:/var/setup:/bin/bash
_on-demand:*:249:249:On Demand Resource Daemon:/var/db/on-demand:/usr/bin/false
_serverdcs:*:251:251:macOS Server Documents Service:/var/empty:/usr/bin/false
_wmmproxy:*:252:252:WMM Proxy:/var/empty:/usr/bin/false
_mobileasset:*:253:253:MobileAsset User:/var/ha:/usr/bin/false
_findmydevice:*:254:254:Find My Device Daemon:/var/db/findmydevice:/usr/bin/false
_datacollectors:*:257:257:DataDetectors:/var/db/datacollectors:/usr/bin/false
_captivagmt:*:258:258:captivagmt:/var/empty:/usr/bin/false
_ctkd:*:259:259:ctkd Account:/var/empty:/usr/bin/false
_applepay:*:260:260:applepay Account:/var/db/applepay:/usr/bin/false
_hid:*:261:261:HID Service User:/var/db/hid:/usr/bin/false
_onddassists:*:262:262:CoreMedia IO Assistants User:/var/db/onddassists:/usr/bin/false
_analyticd:*:263:263:Analytics Daemon:/var/db/analyticd:/usr/bin/false
_fspd:*:265:265:FPS Daemon:/var/db/fpsd:/usr/bin/false
_tined:*:266:266:Time Sync Daemon:/var/db/timed:/usr/bin/false
_reportmemoryexception:*:268:268:Proximity and Ranging Daemon:/var/db/nearbyd:/usr/bin/false
_nearbyd:*:268:268:Proximity and Ranging Daemon:/var/db/nearbyd:/usr/bin/false
_driverkit:*:270:270:DriverKit:/var/empty:/usr/bin/false
_diskimagesd:*:271:271:DiskImages IO Daemon:/var/db/diskimagesd:/usr/bin/false
_logd:*:272:272:Log Daemon:/var/db/diagnostics:/usr/bin/false
_appinstall:*:273:273:App Install Daemon:/var/db/appinstall:/usr/bin/false
_installcoordination:*:274:274:Install Coordination Daemon:/var/db/installcoordination:/usr/bin/false
_dmd:*:275:275:Demo Daemon:/var/empty:/usr/bin/false
_rmd:*:277:277:Remote Management Daemon:/var/db/rmd:/usr/bin/false
_accessoryupdate:*:278:278:Accessory Update Daemon:/var/db/accessoryupdate:/usr/bin/false
_knowledgegraphd:*:279:279:Knowledge Graph Daemon:/var/db/knowledgegraphd:/usr/bin/false
_coreml:*:280:280:CoreML Services:/var/db/coreml:/usr/bin/false
_input:*:281:281:INPUT Server Daemon:/var/empty:/usr/bin/false
_trustd:*:282:282:trustd:/var/empty:/usr/bin/false
_maintenanced:*:283:283:maintenanced:/var/db/maintenanced:/usr/bin/false
_downloadd:*:284:284:Downloadd Daemon:/var/db/downloadd:/usr/bin/false
_notificationproxy:*:285:285:Notification Proxy:/var/empty:/usr/bin/false
_aphidbridge:*:288:288:Apple Virtual Platform HID Bridge:/var/empty:/usr/bin/false
_biome:*:289:289:Biome:/var/db/biome:/usr/bin/false
_backgroundassets:*:291:291:Background Assets Service:/var/empty:/usr/bin/false
_mobilegestalt:*:293:293:MobileGestaltHelper:/var/empty:/usr/bin/false
_audiowd:*:294:294:Audio and MediaExperience Daemon:/var/db/audiowd:/usr/bin/false
_terminusd:*:295:295:Terminus:/var/db/terminus:/usr/bin/false
_neuralengine:*:296:296:AppleNeuralEngine:/var/db/neuralengine:/usr/bin/false
_eligibility:*:297:297:OS Eligibility Daemon:/var/db/eligibility:/usr/bin/false
_systemstatus:*:298:298:SystemStatus Services:/var/empty:/usr/bin/false
_sensed:*:300:300:Always On Sense Daemon:/var/db/sensed:/usr/bin/false
_modelmanager:*:301:301:Model Manager:/var/db/modelmanager:/usr/bin/false
_reportsystemmemory:*:302:302:ReportSystemMemory:/var/empty:/usr/bin/false
_satransparency:*:303:303:Software Transparency Services:/var/db/satransparency:/usr/bin/false
_naturallanguage:*:304:304:Natural Language Services:/var/db/com.apple.naturallanguage:/usr/bin/false
_corespeechd:*:306:306:CoreSpeech Services:/var/empty:/usr/bin/false
_diagnosticservices:*:307:307:Diagnostic Services:/var/empty:/usr/bin/false
_ohd:*:441:441:OHM Daemon:/var/empty:/usr/bin/false
valhokarkuteValhokas-MacBook-Pro Tutedude % head -n 5 /etc/passwd
##
# User Database
#
# Note that this file is consulted directly only when the system is running
# in single-user mode. At other times this information is provided by
valhokarkuteValhokas-MacBook-Pro Tutedude % cat /etc/passwd | tail -n 5
_satransparency:*:303:303:Software Transparency Services:/var/db/satransparency:/usr/bin/false
_naturallanguage:*:304:304:Natural Language Services:/var/db/com.apple.naturallanguage:/usr/bin/false
_corespeechd:*:306:306:CoreSpeech Services:/var/empty:/usr/bin/false
_diagnosticservices:*:307:307:Diagnostic Services:/var/empty:/usr/bin/false
_ohd:*:441:441:OHM Daemon:/var/empty:/usr/bin/false
valhokarkuteValhokas-MacBook-Pro Tutedude %
```

3. Searching for Patterns

Goal: Find lines that contain the word root in /etc/passwd.

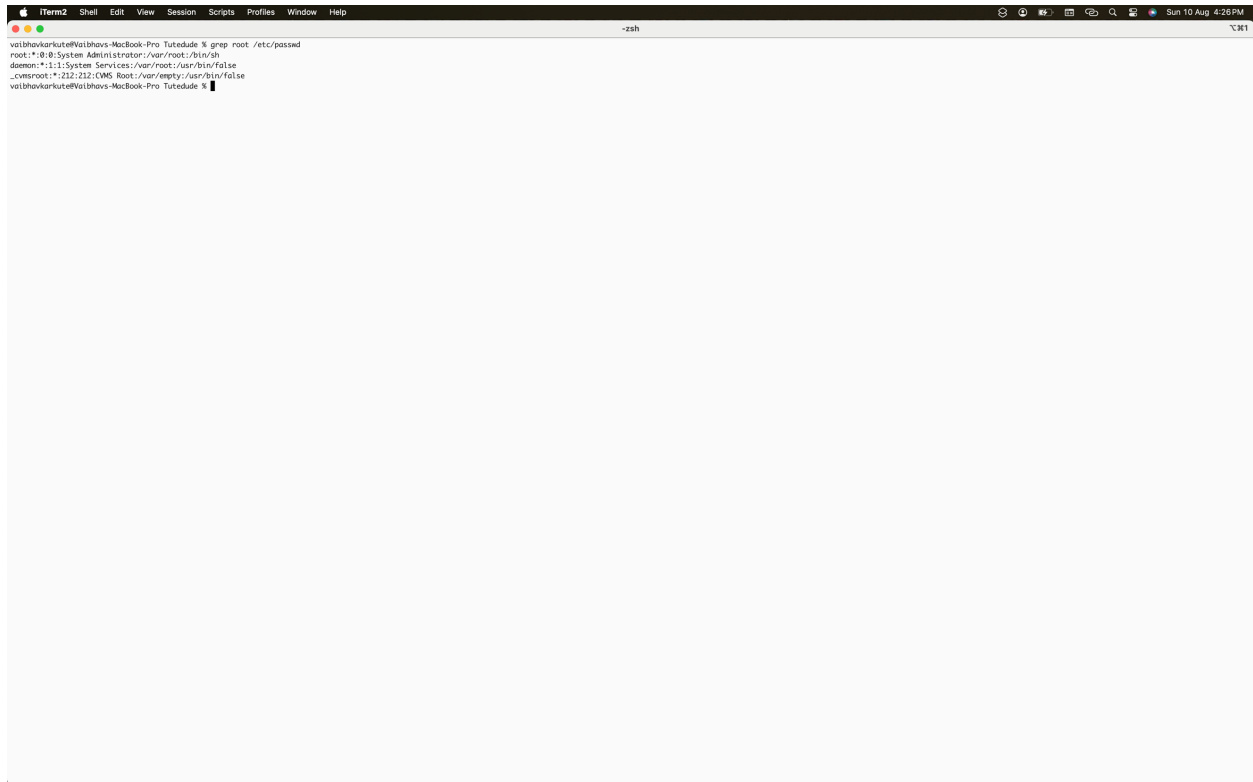
Command:

```
grep 'root' /etc/passwd
```

Why:

- `grep` searches for matching lines. This helps quickly locate root account entries or any other pattern.

Screenshot (evidence):

A screenshot of a terminal window titled 'iTerm2'. The terminal shows the command `grep root /etc/passwd` being executed. The output lists several system accounts: `root:*0:0:System Administrator:/var/root:/bin/sh`, `daemon:*1:1:System Services:/var/root:/usr/bin/false`, and `_overnet:*255:255:OWS_Root:/var/empty:/usr/bin/false`. The prompt indicates the user is 'valbhavkarkute' on a 'MacBook-Pro' machine.

```
iTerm2 Shell Edit View Session Scripts Profiles Window Help
valbhavkarkute@valbhavs-MacBook-Pro Tuteude % grep root /etc/passwd
root:*0:0:System Administrator:/var/root:/bin/sh
daemon:*1:1:System Services:/var/root:/usr/bin/false
_overnet:*255:255:OWS_Root:/var/empty:/usr/bin/false
valbhavkarkute@valbhavs-MacBook-Pro Tuteude %
```

4. Zipping and Unzipping

Goal: Compress `test_dir` into `test_dir.zip` and then extract it into `unzipped_dir`.

Commands:

```
# compress (recursive)
zip -r test_dir.zip test_dir

# create target directory and unzip there
mkdir -p unzipped_dir
unzip test_dir.zip -d unzipped_dir
```

Notes:

- `-r` tells `zip` to include directories recursively. `unzip -d` extracts into the specified directory.

Screenshot (evidence):

```
Item2 Shell Edit View Session Scripts Profiles Window Help
-zsh

._diagnosticservices:*:387:387:Diagnostic Services:/var/empty:/usr/bin/false
._cmd:*:441:441:CMD Domain:/var/empty:/usr/bin/false
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % grep root /etc/passwd
root:*:0:0:System Administrator:/var/root:/bin/sh
daemon:*:1:1:System Services:/var/root:/usr/bin/false
._cvarroot:*:212:212:CMS Root:/var/empty:/usr/bin/false
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % clear
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % grep root /etc/passwd
root:*:0:0:System Administrator:/var/root:/bin/sh
daemon:*:1:1:System Services:/var/root:/usr/bin/false
._cvarroot:*:212:212:CMS Root:/var/empty:/usr/bin/false
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % ls
test_dir
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % zip --version
Copyright (c) 1998-2008 Info-ZIP - Type "zip -l" for software license.
This is Zip 3.0 (July 5th 2008), by Info-ZIP, with modifications by Apple Inc.
Please report bugs to Apple.

Compiled with gcc Apple LLVM 17.0.0 (clang-1700.0.13.5) [internal-os] for Unix (Mac OS X) on Jul 11 2025.

Zip special compilation options:
  USE_32_BIT_TIME      (store Universal Time)
  SYMLINK_SUPPORT      (symbolic links supported)
  LARGE_FILE_SUPPORT   (can read and write large files on file system)
  ZIP64_SUPPORT        (use Zip64 to store large files in archives)
  STORE_UNIX_UIDS_GIDS (store UID/GID sizes/values using new extra field)
  UIDGID_16BIT         (old Unix 16-bit UID/GID extra field also used)
  [encryption, version 2.91 of 85 Jan 2007] (modified for Zip 3)

Encryption notice:
  The encryption code of this program is not copyrighted and is
  put in the public domain. It was originally written in Europe
  and, to the best of our knowledge, can be freely distributed
  in both source and object forms from any country, including
  the USA under License Exception TSU of the U.S. Export
  Administration Regulations (Section 740.13(c)) of 6 June 2002.

Zip environment options:
  ZIP: [none]
  ZIPOPT: [none]
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % zip -r test_dir.zip test_dir
adding: test_dir/ (stored 80%)
adding: test_dir/renamed_example.txt (stored 80%)
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % zip -x test_dir.zip test_dir

zip error: Invalid command arguments (nothing to select from)
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % unzip -x test_dir.zip test_dir
Archive: test_dir.zip
caution: filename not matched: test_dir
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % ls
test_dir
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % rm -rf test_dir
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % unzip -x test_dir.zip test_dir
Archive: test_dir.zip
caution: filename not matched: test_dir
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % ls
test_dir
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % zip -x test_dir.zip test_dir

zip error: Invalid command arguments (nothing to select from)
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % zip -r test_dir.zip test_dir
zip warning: name not matched: test_dir

zip error: Nothing to do! (try: zip -r test_dir.zip . -i test_dir)
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % ls
test_dir.zip
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % unzip test_dir.zip -d test_dir
Archive: test_dir.zip
creating: test_dir/test_dir
extracting: test_dir/test_dir/renamed_example.txt
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude %
```

5. Downloading Files

Goal: Download <https://example.com/sample.txt>.

On macOS you have two common approaches:

A) Using `` (installed by default):****

```
curl -O https://example.com/sample.txt
```

-O saves the file using the remote filename.

B) Using ** (install via Homebrew if you prefer **):

```
# install (if not installed)
```

```
brew install wget
```

```
# then download
```

```
wget https://example.com/sample.txt
```

Notes & troubleshooting (from your screenshot):

- You initially got `zsh: command not found: wget` (means `wget` not installed). The screenshot shows installing `wget` via `brew` and then attempting the download. In your capture the URL returned `404 Not Found` because `example.com/sample.txt` is a placeholder URL — that's expected unless the file actually exists.

Screenshot (evidence):

```
Item2 Shell Edit View Session Scripts Profiles Window Help
-zsh

mcpools: CLI for interacting with MCP servers using both stdio and HTTP transport
media-control: Control and observe media playback from the command-line
msolve: Library for Polynomial System Solving through Algebraic Methods
preflight: Pre-commit re-implemented in Rust
pulp-cli: Command-line interface for Pulp 3
pyrefly: Fast type checker and IDE for Python
rggen: Code generation tool for control and status registers
rulesync: Unified AI rules management CLI tool
secretspec: Declarative secrets management tool
videolchery: Toolkit expanding video processing capabilities
wayback: Archiving tool integrated with various archival services
wgpu-native: Native WebGPU implementation based on wgpu-core

You have 4 outdated formulae installed.

=> Fetching downloads for: wget
=> Downloading https://ghcr.io/v2/homebrew/core/wget/manifests/1.25.0
##### 100.0%
=> Fetching dependencies for wget: libunistring, gettext and libidn2
=> Downloading https://ghcr.io/v2/homebrew/core/libunistring/manifests/1.3
##### 100.0%
=> Fetching libunistring
=> Downloading https://ghcr.io/v2/homebrew/core/libunistring/blobs/sha256:e919f
##### 100.0%
=> Downloading https://ghcr.io/v2/homebrew/core/gettext/manifests/0.26
##### 100.0%
=> Fetching gettext
=> Downloading https://ghcr.io/v2/homebrew/core/gettext/blobs/sha256:87ac92958d
##### 100.0%
=> Downloading https://ghcr.io/v2/homebrew/core/libidn2/manifests/2.3.8
##### 100.0%
=> Fetching libidn2
=> Downloading https://ghcr.io/v2/homebrew/core/libidn2/blobs/sha256:ef2c5796a
##### 100.0%
=> Fetching wget
=> Downloading https://ghcr.io/v2/homebrew/core/wget/blobs/sha256:560877bde7a6
##### 100.0%
=> Installing dependencies for wget: libunistring, gettext and libidn2
=> Installing wget dependency: libunistring
=> Downloading https://ghcr.io/v2/homebrew/core/libunistring/manifests/1.3
Already downloaded: /Users/vaibhavkarkute/Library/Caches/homebrew/downloads/a578da63bc1839c7e217f283da548673ebd899f6e8894d0e79c2eb987c--libunistring-1.3.bottle.manifest.json
=> Pouring libunistring-1.3.sonoma.bottle.tar.gz
  🍺 /usr/local/Cellar/libunistring/1.3: 59 files, 5.9MB
=> Installing wget dependency: gettext
=> Downloading https://ghcr.io/v2/homebrew/core/gettext/manifests/0.26
Already downloaded: /Users/vaibhavkarkute/Library/Caches/homebrew/downloads/d28158ffec84fa757cdeb46750e8e2ed43b7b1ada0d9d725bdad2cfff4d6ed--gettext-0.26.bottle.manifest.json
=> Pouring gettext-0.26.sonoma.bottle.tar.gz
  🍺 /usr/local/Cellar/gettext/0.26: 2,428 files, 27.0MB
=> Installing wget dependency: libidn2
=> Downloading https://ghcr.io/v2/homebrew/core/libidn2/manifests/2.3.8
Already downloaded: /Users/vaibhavkarkute/Library/Caches/homebrew/downloads/f30f50fbde4bf93a71de54d684e482d7d63432656a680097441163c6e5665468--libidn2-2.3.8.bottle.manifest.json
=> Pouring libidn2-2.3.8.sonoma.bottle.tar.gz
  🍺 /usr/local/Cellar/libidn2/2.3.8: 80 files, 838.0KB
=> Installing wget
=> Pouring wget-1.25.0.sonoma.bottle.tar.gz
  🍺 /usr/local/Cellar/wget/1.25.0: 92 files, 4.5MB
=> Running brew cleanup wget...
Disable this behaviour by setting 'HOMEBREW_NO_INSTALL_CLEANUP=1'.
Hide these hints with 'HOMEBREW_NO_INSTALL_HINTS=1' (see 'man brew').
=> No outdated dependents to upgrade!
vaibhavkarkute@Vaibhavs-MacBook-Pro Tutedude % wget https://example.com/sample.txt

--2025-08-10 16:34:03-- https://example.com/sample.txt
Resolving example.com (example.com)... 2600:1406:3a00:21::173e:2605, 2600:1406:3a00:21::173e:2606, 2600:1406:3a00:21::173e:2607, 2600:1406:3a00:21::173e:2608, 2600:1406:3a00:21::173e:2609, 2600:1406:3a00:21::173e:260a, 2600:1406:3a00:21::173e:260b, 2600:1406:3a00:21::173e:260c, 2600:1406:3a00:21::173e:260d, 2600:1406:3a00:21::173e:260e, 2600:1406:3a00:21::173e:260f, 2600:1406:3a00:21::173e:2610, 2600:1406:3a00:21::173e:2611, 2600:1406:3a00:21::173e:2612, 2600:1406:3a00:21::173e:2613, 2600:1406:3a00:21::173e:2614, 2600:1406:3a00:21::173e:2615, 2600:1406:3a00:21::173e:2616, 2600:1406:3a00:21::173e:2617, 2600:1406:3a00:21::173e:2618, 2600:1406:3a00:21::173e:2619, 2600:1406:3a00:21::173e:261a, 2600:1406:3a00:21::173e:261b, 2600:1406:3a00:21::173e:261c, 2600:1406:3a00:21::173e:261d, 2600:1406:3a00:21::173e:261e, 2600:1406:3a00:21::173e:261f, 2600:1406:3a00:21::173e:2620, 2600:1406:3a00:21::173e:2621, 2600:1406:3a00:21::173e:2622, 2600:1406:3a00:21::173e:2623, 2600:1406:3a00:21::173e:2624, 2600:1406:3a00:21::173e:2625, 2600:1406:3a00:21::173e:2626, 2600:1406:3a00:21::173e:2627, 2600:1406:3a00:21::173e:2628, 2600:1406:3a00:21::173e:2629, 2600:1406:3a00:21::173e:262a, 2600:1406:3a00:21::173e:262b, 2600:1406:3a00:21::173e:262c, 2600:1406:3a00:21::173e:262d, 2600:1406:3a00:21::173e:262e, 2600:1406:3a00:21::173e:262f, 2600:1406:3a00:21::173e:2630, 2600:1406:3a00:21::173e:2631, 2600:1406:3a00:21::173e:2632, 2600:1406:3a00:21::173e:2633, 2600:1406:3a00:21::173e:2634, 2600:1406:3a00:21::173e:2635, 2600:1406:3a00:21::173e:2636, 2600:1406:3a00:21::173e:2637, 2600:1406:3a00:21::173e:2638, 2600:1406:3a00:21::173e:2639, 2600:1406:3a00:21::173e:263a, 2600:1406:3a00:21::173e:263b, 2600:1406:3a00:21::173e:263c, 2600:1406:3a00:21::173e:263d, 2600:1406:3a00:21::173e:263e, 2600:1406:3a00:21::173e:263f, 2600:1406:3a00:21::173e:2640, 2600:1406:3a00:21::173e:2641, 2600:1406:3a00:21::173e:2642, 2600:1406:3a00:21::173e:2643, 2600:1406:3a00:21::173e:2644, 2600:1406:3a00:21::173e:2645, 2600:1406:3a00:21::173e:2646, 2600:1406:3a00:21::173e:2647, 2600:1406:3a00:21::173e:2648, 2600:1406:3a00:21::173e:2649, 2600:1406:3a00:21::173e:264a, 2600:1406:3a00:21::173e:264b, 2600:1406:3a00:21::173e:264c, 2600:1406:3a00:21::173e:264d, 2600:1406:3a00:21::173e:264e, 2600:1406:3a00:21::173e:264f, 2600:1406:3a00:21::173e:2650, 2600:1406:3a00:21::173e:2651, 2600:1406:3a00:21::173e:2652, 2600:1406:3a00:21::173e:2653, 2600:1406:3a00:21::173e:2654, 2600:1406:3a00:21::173e:2655, 2600:1406:3a00:21::173e:2656, 2600:1406:3a00:21::173e:2657, 2600:1406:3a00:21::173e:2658, 2600:1406:3a00:21::173e:2659, 2600:1406:3a00:21::173e:265a, 2600:1406:3a00:21::173e:265b, 2600:1406:3a00:21::173e:265c, 2600:1406:3a00:21::173e:265d, 2600:1406:3a00:21::173e:265e, 2600:1406:3a00:21::173e:265f, 2600:1406:3a00:21::173e:2660, 2600:1406:3a00:21::173e:2661, 2600:1406:3a00:21::173e:2662, 2600:1406:3a00:21::173e:2663, 2600:1406:3a00:21::173e:2664, 2600:1406:3a00:21::173e:2665, 2600:1406:3a00:21::173e:2666, 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```

```

❯ iTerm2 Shell Edit View Session Scripts Profiles Window Help
~zsh
secretapec: Declarative secrets management tool
videolachery: Toolkit expanding video processing capabilities
waybackr: Archiving tool integrated with various archival services
wgpu-native: Native WebGPU implementation based on wgpu-core

You have 4 outdated formulae installed.

--> Fetching downloads for: wget
--> Downloading https://ghcr.io/v2/homebrew/core/wget/manifests/1.25.0
##### 100.0%
--> Fetching dependencies for wget: libunistring, gettext and libidn2
--> Downloading https://ghcr.io/v2/homebrew/core/libunistring/manifests/1.3
##### 100.0%
--> Fetching libunistring
--> Downloading https://ghcr.io/v2/homebrew/core/libunistring/blobs/sha256:e919f
##### 100.0%
--> Downloading https://ghcr.io/v2/homebrew/core/gettext/manifests/0.26
##### 100.0%
--> Fetching gettext
--> Downloading https://ghcr.io/v2/homebrew/core/gettext/blobs/sha256:87ac92958d
##### 100.0%
--> Downloading https://ghcr.io/v2/homebrew/core/libidn2/manifests/2.3.8
##### 100.0%
--> Fetching libidn2
--> Downloading https://ghcr.io/v2/homebrew/core/libidn2/blobs/sha256:ef2c85790a
##### 100.0%
--> Fetching wget
--> Downloading https://ghcr.io/v2/homebrew/core/wget/blobs/sha256:56580778de7a6
##### 100.0%
--> Installing dependencies for wget: libunistring, gettext and libidn2
--> Installing wget dependency: libunistring
--> Downloading https://ghcr.io/v2/homebrew/core/libunistring/manifests/1.3
Already downloaded: /Users/vaibhavkarkute/Library/Caches/Homebrew/downloads/628158ffec84fa757cd8eb4675b8e2ed43b7917ada9d72f5bdad2c44cded--gettext-0.26.bottle.manifest.json
--> Pouring libunistring-1.3.somoma.bottle.tar.gz
📦 /usr/local/Cellar/libunistring/1.3: 59 files, 5.9MB
--> Installing wget dependency: gettext
--> Downloading https://ghcr.io/v2/homebrew/core/gettext/manifests/0.26
Already downloaded: /Users/vaibhavkarkute/Library/Caches/Homebrew/downloads/628158ffec84fa757cd8eb4675b8e2ed43b7917ada9d72f5bdad2c44cded--gettext-0.26.bottle.manifest.json
--> Pouring gettext-0.26.somoma.bottle.tar.gz
📦 /usr/local/Cellar/gettext/0.26: 2,428 files, 27.0MB
--> Installing wget dependency: libidn2
--> Downloading https://ghcr.io/v2/homebrew/core/libidn2/manifests/2.3.8
Already downloaded: /Users/vaibhavkarkute/Library/Caches/Homebrew/downloads/3f8f58f8de4bf3a71de54d684e482d7d3432656d88b97441163c6e5665468--libidn2-2.3.8.bottle.manifest.json
--> Pouring libidn2-2.3.8.somoma.bottle.tar.gz
📦 /usr/local/Cellar/libidn2/2.3.8: 88 files, 838.2KB
--> Installing wget
--> Pouring wget-1.25.0.somoma.bottle.tar.gz
📦 /usr/local/Cellar/wget/1.25.0: 92 files, 4.5MB
--> Running 'brew cleanup wget'...
Disable this behaviour by setting 'HOMEBREW_NO_INSTALL_CLEANUP=1'.
Hide these hints with 'HOMEBREW_NO_INSTALL_HINTS=1' (see 'man brew').
--> No outdated dependents to upgrade!
vaibhavkarkute@Vaibhavs-MacBook-Pro TuteDude % wget https://example.com/sample.txt
--2025-08-10 16:34:03-- https://example.com/sample.txt
Resolving example.com (example.com)... 2080:1406:3a00:21::173a:2a65, 2080:1406:3a00:21::173a:2a66, 2080:1406:bc00:53::b81e:94c8, ...
Connecting to example.com (example.com)[2080:1406:3a00:21::173a:2a65]:443... failed: Operation timed out.
Connecting to example.com (example.com)[2080:1406:3a00:21::173a:2a66]:443... failed: Operation timed out.
Connecting to example.com (example.com)[2080:1406:bc00:53::b81e:94c8]:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2025-08-10 16:34:12 ERROR 404: Not Found.

vaibhavkarkute@Vaibhavs-MacBook-Pro TuteDude % touch secure.txt
vaibhavkarkute@Vaibhavs-MacBook-Pro TuteDude % ls
secure.txt  test_dir
vaibhavkarkute@Vaibhavs-MacBook-Pro TuteDude % chmod 400 secure.txt
vaibhavkarkute@Vaibhavs-MacBook-Pro TuteDude % ls
secure.txt  test_dir
vaibhavkarkute@Vaibhavs-MacBook-Pro TuteDude % ls -l secure.txt
-r----- 1 vaibhavkarkute staff 0 10 Aug 16:34 secure.txt
vaibhavkarkute@Vaibhavs-MacBook-Pro TuteDude %
```

7. Working with Environment Variables

Goal: Set MY_VAR to "Hello, Linux!".

Commands:

```
export MY_VAR="Hello, Linux!"
echo $MY_VAR
```

Make it persistent (zsh on macOS):

```
echo 'export MY_VAR="Hello, Linux!"' >> ~/.zshrc
source ~/.zshrc
```

Screenshot (evidence):

```
iTerm2  Shell  Edit  View  Session  Scripts  Profiles  Window  Help
-zsh
vaibhavkarte@Vaibhavs-MacBook-Pro Tutedude % export MY_VAR="Hello, Linux!"

dquote> q
dquote>
vaibhavkarte@Vaibhavs-MacBook-Pro Tutedude % echo $MY_VAR

vaibhavkarte@Vaibhavs-MacBook-Pro Tutedude % export MY_VAR="Hello, Linux"
vaibhavkarte@Vaibhavs-MacBook-Pro Tutedude % echo $MY_VAR

Hello, Linux
vaibhavkarte@Vaibhavs-MacBook-Pro Tutedude % █
```